

LECTURE SHEET

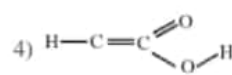
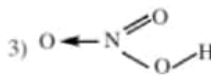
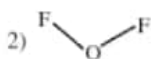
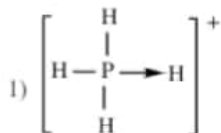
EXERCISE-I

(Lewis Theory, Ionic bond, Fajan rules, Covalent bond (overlapping part))

LEVEL-I (MAIN)

Straight Objective Type Questions

- The correct order of the lattice energies of the following ionic compounds is
 1) $\text{NaCl} > \text{MgBr}_2 > \text{CaO} > \text{Al}_2\text{O}_3$ 2) $\text{Al}_2\text{O}_3 > \text{MgBr}_2 > \text{CaO} > \text{NaCl}$
 3) $\text{MgBr}_2 > \text{Al}_2\text{O}_3 > \text{CaO} > \text{NaCl}$ 4) $\text{Al}_2\text{O}_3 > \text{CaO} > \text{MgBr}_2 > \text{NaCl}$
- In which of the following the hydration energy is higher than the lattice energy ?
 1) MgSO_4 2) RaSO_4 3) SrSO_4 4) BaSO_4
- A molecule which cannot exist theoretically is
 1) SF_4 2) OF_2 3) OF_4 4) O_2F_2
- Among LiCl , BeCl_2 , BCl_3 and CCl_4 , the covalent bond character follows the order
 1) $\text{LiCl} > \text{BeCl}_2 > \text{BCl}_3 > \text{CCl}_4$ 2) $\text{LiCl} < \text{BeCl}_2 < \text{BCl}_3 < \text{CCl}_4$
 3) $\text{LiCl} > \text{BeCl}_2 > \text{CCl}_4 > \text{BCl}_3$ 4) $\text{LiCl} > \text{BeCl}_2 < \text{BCl}_3 > \text{CCl}_4$
- The correct order of decreasing polarisability of ion is
 1) $\text{Cl}^- > \text{Br}^- > \text{I}^- > \text{F}^-$ 2) $\text{F}^- > \text{I}^- > \text{Br}^- > \text{Cl}^-$ 3) $\text{I}^- > \text{Br}^- > \text{Cl}^- > \text{F}^-$ 4) $\text{F}^- > \text{Cl}^- > \text{Br}^- > \text{I}^-$
- Which ion has a higher polarising power ?
 1) Mg^{2+} 2) Al^{3+} 3) Ca^{2+} 4) Na^+
- The correct order of increasing covalent character is
 1) $\text{NaCl} < \text{LiCl} < \text{BeCl}_2$ 2) $\text{BeCl}_2 < \text{NaCl} < \text{LiCl}$
 3) $\text{BeCl}_2 < \text{LiCl} < \text{NaCl}$ 4) $\text{LiCl} < \text{NaCl} < \text{BeCl}_2$
- The pair with more ionic nature and less m.pt. respectively among lithium halides
 1) LiF , LiI 2) LiCl , LiF 3) LiBr , LiI 4) LiF , LiCl
- One would expect proton to have very large :
 1) ionisation potential 2) radius 3) charge 4) hydration energy
- Octet rule is not valid for the molecule
 1) CO_2 2) NO 3) O_2 4) PCl_3
- Hypervalent compound is
 1) IF_7 2) NH_3 3) BeF_2 4) CH_4
- Which formulae does not correctly represents the bonding capacity of the atom involved ?



13. Which has a giant covalent structure ?
 1) PbO_2 2) SiO_2 3) NaCl 4) AlCl_3
14. O_2^{2-} is isoelectronic with
 1) H_2 2) N_2 3) F_2 4) S
15. How many bonded electron pairs are present in IF_7 molecule ?
 1) 6 2) 7 3) 5 4) 8

LEVEL-II (ADVANCED)

Straight Objective Type Questions

- Valency of an element indicates
 a) combining power with hydrogen b) acidity
 c) electrons in the outermost orbit d) basicity
- Among atoms of a molecule, there exists
 a) only attractive forces b) only repulsive forces
 c) both attractive and repulsive forces d) neither attractive nor repulsive forces
- In ammonium ion the covalency of nitrogen is
 a) 3 b) 4 c) 2 d) 5
- Most energetic species among the following is
 a) H_2 b) Ne c) F d) F_2
- Which of the following molecules contain one lone pair of electrons on the central atom?
 a) CH_4 b) PH_3 c) CCl_4 d) H_2S
- Which of the following molecule does not obey the octet rule and also has lone pair on the central atom.
 a) CCl_4 b) PCl_3 c) NH_3 d) SiCl_4
- The total number of electrons that take part in forming bonds in a C_2H_4 molecule are
 a) 12 b) 14 c) 6 d) 10
- The formal charges on the three oxygen atoms in O_3 molecule are
 a) 0, 0, 0 b) 0, 0, -1 c) 0, 0 + 1 d) 0, +1, -1
- Which of the following is easily formed?
 a) Calcium chloride b) Calcium bromide c) Potassium chloride d) Potassium bromide
- Among the compounds NaCl , KCl , RbCl and CsCl , the one with greatest ionic character is
 a) NaCl b) KCl c) CsCl d) RbCl
- Molten sodium chloride conducts electricity due to the presence of
 a) Free electrons b) Free ions c) Free molecules d) Free atoms
- The total number of Na^+ ions present per unit cell of NaCl is
 a) 1 b) 6 c) 8 d) 4
- The number of ion pairs that constitute one unit cell of CsCl
 a) 4 b) 2 c) 8 d) 1

14. How many unit cells are present in a cube-shaped ideal crystal of NaCl of mass 1g?
(atomic masses : Na = 23, Cl = 35.5)
- a) 1.7×10^{21} unit cells b) 2.57×10^{21} unit cells
c) 5.14×10^{21} unit cells d) 1.28×10^{21} unit cells
15. Which of the following is not involved in Born-Haber cycle?
- a) Sublimation energy b) Ionisation potential
c) Lattice energy d) Electronegativity
16. What is the crystal structure of CsCl?
- a) Body centered cubic b) Face centered cubic
c) Tetrahedral d) Octahedral
17. Which of the following is a favourable factor for cation formation?
- a) High electronegativity b) High electron affinity
c) Low ionisation potential d) Smaller atomic size
18. Two elements 'X' and 'Y' have the following configuration X = $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$, Y = $1s^2 2s^2 2p^6 3s^2 3p^5$. The compound formed by the combination of 'X' and 'Y' will be
- a) XY_2 b) X_5Y_2 c) X_2Y_5 d) XY_5
19. From the electronic structure of four elements predict the one with greater tendency to form electrovalent bond is
- a) $1s^2$ b) $1s^2, 2s^2, 2p^2$ c) $1s^2, 2s^2, 2p^5$ d) $1s^2, 2s^2, 2p^6$
20. Which of the following is favourable condition for the formation of ionic bond?
- a) Small cation with small charge b) Small anion with large charge
c) Large difference in the electronegativity d) Small cation with high charge

More than One correct answer Type Questions

21. The bond between the atoms may be formed by
- a) Transfer of electrons b) Sharing of electrons
c) Transfer of protons d) Transfer of neutrons
22. Amongst, the species with formal charge '-1' on the central atom
- a) BF_4^- b) ICl_3 c) CO_3^{2-} d) I_3^-
23. Two elements have electronegativity of 1.2 and 3.0 bond formed between them would be
- a) Ionic b) Polar covalent c) Co-ordinate d) Metallic
24. Among the following which hydrides are covalent
- a) NH_3 b) B_2H_6 c) NaH d) CaH_2
25. Most favourable condition for ionic bonding are
- a) Low charge on ions, large cations, small anions
b) Low charge on ions, large cations, large anions
c) High charge on ions, small cations, large anions
d) High charge on ions, large cations, small anions

26. Which of the following is not correct statement?
- Ionic compound are electrically neutral
 - Boiling point of ionic compound is more than covalent compound
 - Melting point of covalent compound is more than ionic compound
 - Ionic compound are soluble in polar solvent

Linked Comprehension Type Questions

Passage-I :

The Chemical bond formed due to electron transfer is called ionic bond or electrovalent bond. Ionic bond will be formed more easily between the elements with low ionization potential and high electron affinity. Energy changes involved during the formation of ionic compound can be calculated by Born-Haber cycle. Lattice enthalpy changes are directly proportional to the stability of ionic compound.

27. Which of the following has electrovalent bond?
- HCl
 - AlF_3
 - CH_4
 - BeCl_2
28. Born-Haber cycle is based on
- Faraday's law
 - Gay-Lussac's law
 - Emeton's law
 - Hess's law
29. Most stable ionic compound among the following is
- Li_2O
 - MgO
 - Cs_2O
 - KI

Passage-II :

Fajan rules :

- The smaller the cation, the higher is its polarizing power.
 - Cations with pseudo noble gas configuration ($ns^2np^6nd^{10}$) having relatively high polarizing power than those with noble gas configuration (ns^2np^6).
 - The larger the size of the anion, the higher is its polarisability.
30. Among the following LiCl , BeCl_2 , BCl_3 , CCl_4 which will have the lowest melting point and the highest solubility in water.
- BCl_3 , BeCl_2
 - LiCl , CCl_4
 - BeCl_2 , BCl_3
 - CCl_4 , LiCl
31. The correct order of polarisability is I^- , Br^- , Cl^- , F^-
- $\text{I}^- > \text{Br}^- > \text{Cl}^- > \text{F}^-$
 - $\text{I}^- > \text{Br}^- = \text{Cl}^- > \text{F}^-$
 - $\text{I}^- = \text{Br}^- = \text{Cl}^- > \text{F}^-$
 - $\text{I}^- = \text{Br}^- < \text{Cl}^- = \text{F}^-$
32. The ionic conductance of which of the following is the highest ?
- Li^+ (aq)
 - Na^+ (aq)
 - K^+ (aq)
 - Cs^+ (aq)

Matrix Matching Type Questions

33. **Column-I**
- Highest polarizing power
 - Highest polarisability
 - Ion with pseudo inert gas configuration
 - Least polarisability
- Column-II**
- F^-
 - Be^{2+}
 - Zn^{2+}
 - I^-

Integer Type Questions

34. How many of the following compounds having odd number of electrons:
 Cs_2O , MgO , Cl_2O , ClO_2 , NO , NO_2 , N_2O_4 , N_2O_5 , CaOCl_2
35. How many of the following compounds not having complete octet:
 ClO_2 , NO , NO_2 , SF_6 , H_2SO_4 , BF_3 , BeCl_2
36. What is the formal charge on each O-atoms in the ion $[\ddot{\text{O}}=\text{N}=\ddot{\text{O}}]^+$
37. The number of moles of ions present in one mole of unit cells of NaCl
38. Calculate the coordinate number in an ionic crystal, when the radius ratio of cation to anion is 0.3

EXERCISE-II*(VSEPR Theory, Valency bond theory, Hybridization)***LEVEL-I (MAIN)**Straight Objective Type Questions

- In HCHO , there are X non-bonding electron pairs, Y σ -bonds and Z π -bonds, X, Y and Z are
 1) 1, 1, 3 2) 2, 3, 1 3) 1, 2, 3 4) 1, 3, 2
- For compounds.
 A) Tetracyanoethene B) Carbon dioxide C) Benzene D) 1,3-butadiene
 Ratio of σ and π bonds is in order
 1) $A = B < C < D$ 2) $A = B < D < C$ 3) $A = B = C = D$ 4) $C < D < A < B$
- In allene structure, three carbon atoms are joined by
 1) Three σ - and three π - bonds 2) two σ - bonds and one π - bond
 3) two σ - and two π - bonds 4) three π - bonds only
- Among the following compounds the one that is polar and has the central atom with sp^3 hybridisation is
 1) H_2CO_3 2) SiF_4 3) BF_3 4) HClO_2
- The pair in which the two species are isostructural ?
 1) SiF_4 and SF_4 2) IO_3^- and XeO_3 3) BH_4^- and NH_4^+ 4) PF_6^- and SF_6
- In NO_3^- ion, the number of bond pair and lone pair of electrons on N-atom are
 1) 2, 2 2) 3, 1 3) 1, 3 4) 4, 0
- In which of the following species the hybrid state of the central atom is same ?
 1) SO_2 , SO_3 2) SO_3 , SO_4^{2-} 3) SO_2 , SO_3^{2-} 4) CH_4 , HCOOH
- Among the following which species has same number of σ and π bonds ?
 1) C_7H_8 2) $\text{C}_2(\text{CN})_4$ 3) C_2H_4 4) $\text{HC} \equiv \text{CH}$
- The pair having similar geometry is
 1) BF_3 , NF_3 2) BF_3 , AlF_3 3) BeF_2 , H_2O 4) BCl_3 , PCl_3
- The shape of ClO_3^- according to VSEPR model is
 1) planar triangle 2) pyramidal 3) tetrahedral 4) square planar

11. Number of π -bonds in naphthalene is
 1) 6 2) 3 3) 4 4) 5
12. Which pair is isostructural and possesses same number of lone pair of electron on central atom ?
 1) IF_5 and XeOF_4 2) NH_3 and ClO_3^- 3) SnCl_4 and ClO_4^- 4) AlCl_3 and SO_2
13. H-B-H bond angle in BH_4^- is
 1) 180° 2) 120° 3) 109° 4) 90°
14. CO_2 has the same geometry as
 A) HgCl_2 B) NO_2 C) SnCl_4 D) C_2H_2
 1) A and C 2) B and D 3) A and D 4) C and D
15. In PH_3 and $\text{P}(\text{CH}_3)_3$, the atomic/hybrid orbitals of P which take part in the formation of P-H bond, are
 1) sp^3 , sp^3 2) p, p 3) p, sp^3 4) sp^3 , p

LEVEL-II (ADVANCED)

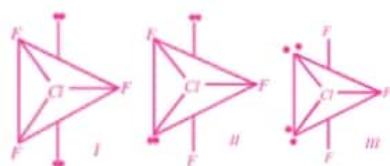
Straight Objective Type Questions

1. Most important concept of valence bond theory is
 a) Overlap of atomic orbitals results in the bond b) Sharing of odd number of electrons for bonding
 c) Sharing of electrons follow the octet rule d) Transfer of electrons follow the octet rule
2. According to valence bond theory, water molecule has
 a) Two dative bonds and bond angle 90° b) Two covalent bonds and bond angle 90°
 c) Two dative bonds and bond angle 104.5° d) Two covalent bonds and bond angle 104.5°
3. Sigma bonds are :
 a) Cylindrically symmetrical about the bond axis b) Perpendicular to the bond axis
 c) asymmetrical about the bond axis d) None of the above
4. Which of the following is called negative overlap?
 a) b) c) d)
5. Which one or more among the following involve(s) $\text{p}\pi-\text{d}\pi$ bonding?
 a) $(\text{SiH}_3)_3\text{N:}$ b) $(\text{CH}_3)_3\text{N:}$ c) $:\text{CCl}_3^-$ d) $:\text{CF}_3^-$
6. The ion that is iso-electronic with CO
 a) O_2^+ b) CN^- c) O_2^- d) N_2^+
7. In N_2 molecule the atoms are bonded by the bonds
 a) 1σ , 2π b) 1σ , 1π c) 2σ , 1π d) 3σ , 2π
8. Two carbon atoms in which of the following have more number of shared electrons
 a) Benzene b) Acetylene c) Ethane d) Ethylene
9. A molecule AX_2 has two lone pairs over A. Its shape is
 a) Tetrahedral b) Pyramidal c) Angular d) Linear

10. The geometry of H_3O^+ ion is
 a) Planar b) Triangular c) Pyramidal d) Tetrahedral
11. The shape of AB_3E is
 a) Pyramidal b) Tetrahedral c) Angular d) Linear
12. The hybridization involved in acetylene is
 a) sp b) sp^2 c) sp^3 d) dsp^2
13. Hybridization produces a set of orbitals which are
 a) Parallel b) Perpendicular c) Equivalent d) Unequivalent
14. Maximum number of planar atoms in SF_6 molecule
 a) 5 b) 4 c) 6 d) 7
15. Which hybridisation is found in NH_3 and H_2O
 a) sp^3 b) dsp^2 c) sp d) sp^2
16. Octahedral shape is due to the hybridisation
 a) sp^3d b) sp^3d^2 c) sp^3 d) sp
17. The correct order of bond angles (smallest first) in H_2S , NH_3 , BF_3 and SiH_4 is
 a) $\text{H}_2\text{S} < \text{NH}_3 < \text{SiH}_4 < \text{BF}_3$ b) $\text{NH}_3 < \text{H}_2\text{S} < \text{SiH}_4 < \text{BF}_3$
 c) $\text{H}_2\text{S} < \text{SiH}_4 < \text{NH}_3 < \text{BF}_3$ d) $\text{NH}_3 < \text{H}_2\text{S} < \text{BF}_3 < \text{SiH}_4$

More than One correct answer Type Questions

18. Which statement(s) is(are) correct ?
 a) A sigma bond is stronger than π -bond
 b) A sigma bond is weaker than π -bond
 c) Hydrogen bonding is weaker than covalent bonding
 d) A triple bond is weaker than double bond
19. Geometry of the molecule is distorted by VSEPR in
 a) H_2O b) NH_3 c) N_3H d) XeF_2
20. Which of the following shows iso-structural species ?
 a) NH_4^+ and AlCl_4^- b) CH_3^- and H_3O^+
 c) SO_4^{2-} , PO_4^{3-} and BF_4^- d) SiCl_4 and CO_2
21. In ClF_3 the central chlorine atom is sp^3d hybridised and theoretically three structures are possible.



Which of the following is true.

- a) All the 3 structure are equally stable b) Structure III is most stable
 c) In structure III there are four 90° lp – bp repulsions and no 90° lp – lp repulsions.
 d) In structure I there are 6 lp – lp repulsions at 90°

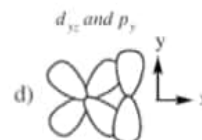
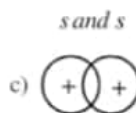
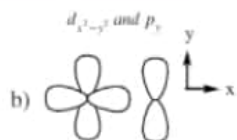
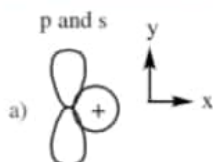
22. The correct statement about sulphur hexa fluoride include
 a) There are 12 F – S – F 90° bond angles
 b) S in SF_6 has an expanded octet
 c) In SF_6 all fluorines are in the same plane
 d) SF_6 has a distorted octahedral geometry
23. The linear structure is assumed by :
 a) SnCl_2
 b) CO_2
 c) NO_2^+
 d) CS_2
24. Which combination of the compounds and their geometry are correct ?
 a) HgCl_2 - linear
 b) ClF_3 - V-shaped
 c) ClF_3 - T-shaped
 d) ICl_4^- - square planar
25. Which of the following have a bond angle smaller than tetrahedral bond ?
 a) NH_4^+
 b) SiH_4
 c) NH_3
 d) H_2O
26. In which of the following molecules, vacant orbitals take part in hybridization?
 a) B_2H_6
 b) Al_2Cl_6
 c) H_3PO_3
 d) H_3BO_3
27. Which of the following molecules have (–O–O–) bond, i.e., peroxide bond?
 a) $\text{H}_2\text{S}_2\text{O}_8$
 b) $\text{H}_2\text{S}_2\text{O}_7$
 c) H_2SO_5
 d) $\text{H}_2\text{S}_2\text{O}_6$

Linked Comprehension Type Questions

Passage-I :

When half-filled valence orbital of one atom overlaps with half-filled valence orbital of other atom, a covalent bond is formed. Head on or axial overlap leads to the formation of sigma bond. Sidewise or lateral overlap of atomic orbitals (p) give rise to the formation of π -bond. Similarly side wise overlap between d-orbitals can lead to the formation of δ -bond. In case of π -bond, the probability of finding the shared electron cloud is zero along the line joining two nuclei while the probability of finding the shared electron cloud is maximum along the inter nuclear axis.

28. If x-axis is the internuclear axis then which of the following combination will result into the formation of π -bond?
 a) $s + 2p_x$
 b) $2p_x + 2p_x$
 c) $2p_y + 2p_y$
 d) $s + s$
29. The strength of bonds by s-s, p-p, s-p overlap is in the order (excluding 1s):
 a) $s-s < s-p < p-p$
 b) $s-s < p-p < s-p$
 c) $s-p < s-s < p-p$
 d) $p-p < s-s < s-p$
30. If the internuclear axis is x-axis, which of the following will not give non bonding combination?



Passage-II :

Most of the polyatomic molecules except a few such as CO_2 and CS_2 which are linear or angular with a bond angle generally somewhat greater than 90° . A bond angle is defined as the angle between the direction of two covalent bonds. Since the atoms in molecules are in constant motion with respect to each other, they are not expected to have a fixed value of bond angle.

31. Which of the following having highest bond angle ?
 a) H_2S
 b) H_2Se
 c) H_2O
 d) All have same bond angle

32. In P_4 molecule phosphorus atoms are tetrahedrally arranged. The bond angle P-P-P in the molecule is
 a) 108° b) 120° c) 60° d) 180°
33. Which of the following hybridisation may have more than one type of bond angle?
 a) sp^2 b) sp^3 c) sp^3d d) sp^3d^2

Matrix Matching Type Questions**34. Column-I**

- A) Electron deficient
 B) Odd electron molecule
 C) Expansion of octet
 D) T shaped molecule

Column-II

- P) ClF_3
 Q) $BeCl_2$
 R) BF_3
 S) NO

Integer Type Questions

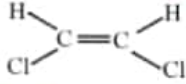
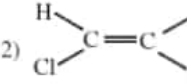
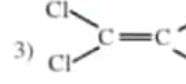
35. Based on VSEPR theory, the number of 90° F-Br-F angles in BrF_5 is
 36. What are the total number of hybrid orbitals present on Xe in XeF_6 ?
 37. Total number of covalent bonds between carbon and carbon atoms in C_3O_2 molecule is
 38. The number of covalent bonds in XeO_4 molecule is
 39. In SF_4 , S atom has dsp^x hybridisation. What is the value of x ?
 40. Give the number of compounds with two lone pairs each among XeF_5^- , BrF_3 , XeF_2 , H_3S^+ is ____
 41. The number of π bonds in acrylonitrile molecule is
 42. How many equatorial bonds are there in PCl_5 molecule?
 43. The number of $(p\pi - d\pi)$ bonds in XeO_4 is _____.
 44. How many 'sp' hybrid orbitals are there in allene C_3H_4 ?

EXERCISE-III

(Bond parameters (Resonance, Dipole moment, Bond length, Bond angles, Bond energies),
 Coordinate covalent bond)

LEVEL-I (MAIN)**Straight Objective Type Questions**

1. Among the following which has resonating structure ?
 1) O_3 2) PCl_5 3) SF_6 4) IF_7
2. The ONO angle is maximum in
 1) NO_3^- 2) NO_2 3) NO_2^- 4) NO_2^+
3. The correct increasing bond angles order is
 1) $BF_3 < NF_3 < PF_3 > ClF_3$ 2) $ClF_3 < PF_3 < NF_3 < BF_3$
 3) $BF_3 = NF_3 < PF_3 < ClF_3$ 4) $BF_3 < NF_3 < PF_3 < ClF_3$
4. The correct order of increasing bond angles is :
 1) $PF_3 < PCl_3 < PBr_3 < PI_3$ 2) $PF_3 < PBr_3 < PCl_3 < PI_3$
 3) $PI_3 < PBr_3 < PCl_3 < PF_3$ 4) $PF_3 > PCl_3 < PBr_3 < PI_3$

5. The correct order for bond angles is
 1) $\text{NO}_2^+ > \text{NO}_2 > \text{NO}_2^-$ 2) $\text{NO}_2^+ > \text{NO}_2^- > \text{NO}_2$ 3) $\text{NO}_2 > \text{NO}_2^- > \text{NO}_2^+$ 4) $\text{NO}_2^- > \text{NO}_2 > \text{NO}_2^+$
6. The correct order of bond angles is
 1) $\text{NH}_4^+ > \text{NH}_3 > \text{NH}_2^-$ 2) $\text{NH}_3 > \text{NH}_4^+ > \text{NH}_2^-$ 3) $\text{NH}_4^+ > \text{NH}_2^- > \text{NH}_3$ 4) $\text{NH}_3 > \text{NH}_2^- > \text{NH}_4^+$
7. In which of the following bond angle is maximum ?
 1) NH_3 2) NH_4^+ 3) PCl_5 4) SCl_2
8. Dative bond is present in
 1) O_3 2) NH_3 3) BaCl_2 4) BF_3
9. Which of the following does not contain co-ordinate bond ?
 1) BH_4^- 2) NH_2^- 3) SO_3 4) H_3^+O
10. Which have zero dipole moment ?
 1) 1,1-Dichloroethene 2) Cis-1,2-dichloroethane
 3) Trans-1,2 dichloroethene 4) 1,2-Dichlorobenzene
11. Select the correct relation
 1) $\mu_{\text{NH}_3} = \mu_{\text{NF}_3}$ 2) $\mu_{\text{NH}_3} > \mu_{\text{NF}_3}$ 3) $\mu_{\text{NH}_3} < \mu_{\text{NF}_3}$ 4) can't compared
12. Select the isomers given below, which have non-zero dipole moment ?
 1)  2)  3)  4) All of these
13. Select the correct relation :
 1) $\mu_{\text{CH}_3\text{OH}} = \mu_{\text{CH}_3\text{SH}}$ 2) $\mu_{\text{CH}_3\text{OH}} > \mu_{\text{CH}_3\text{SH}}$ 3) $\mu_{\text{CH}_3\text{OH}} < \mu_{\text{CH}_3\text{SH}}$ 4) Can't compared
14. Dipole moment is highest for
 1) Cl_4 2) CH_4 3) CHF_3 4) CCl_4
15. Dipole moment is shown by
 1) 1,4-dichlorobenzene 2) cis-1,2-dichloro-1-butene
 3) trans-2,3-dichloro-2-butene 4) trans-2-butene
16. Which of the following molecules does not possess a permanent electric dipole moment ?
 1) H_2S 2) SO_2 3) SO_3 4) CS_2
17. Which of the following would have permanent dipole moment ?
 1) SF_4 2) XeF_4 3) SiF_4 4) BF_3
18. Which bond angle ' θ ' would result in maximum dipole moment for the triatomic molecule XY_2 ?
 1) $\theta = 90^\circ$ 2) $\theta = 120^\circ$ 3) $\theta = 150^\circ$ 4) $\theta = 180^\circ$
19. The dipole moment of NF_3 is very much less compared to that of NH_3 because
 1) the size of N-atom is much less than that of H-atom
 2) F-atom is more electronegative than N-atom, where as H-atom is less electronegative than N-atom
 3) unshared electron pair is not present in NF_3
 4) number of lone pairs in NF_3 is much greater than in NF_3

20. PH_3 and BF_3 form an adduct readily because they form
- 1) A coordinate bond 2) A covalent bond 3) An ionic bond 4) A hydrogen bond
21. Which of the following does not contain coordinate covalent bond ?
- 1) NH_4^+ 2) H_3O^+ 3) CH_3^- 4) $[\text{Ag}(\text{CN})_2]^+$

LEVEL-II (ADVANCED)

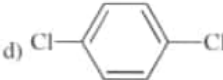
Straight Objective Type Questions

- The bond length of $\text{C} \equiv \text{O}$ bond in carbon monoxide is $1.20 \text{ \AA}$ and in carbon dioxide it is $1.34 \text{ \AA}$. Then the $\text{C} = \text{O}$ bond length in CO_3^{2-} will be
 - $0.95 \text{ \AA}$
 - $1.50 \text{ \AA}$
 - $1.29 \text{ \AA}$
 - $1.34 \text{ \AA}$
- The $\text{C} - \text{C}$ bond length is $1.54 \text{ \AA}$, $\text{C} = \text{C}$ bond length is $1.33 \text{ \AA}$. What is the circumference of benzene ring?
 - $(3 \times 1.54 + 3 \times 1.33) \text{ \AA}$
 - $(4 \times 1.54 + 2 \times 1.33) \text{ \AA}$
 - $(6 \times 1.4) \text{ \AA}$ (1.4 is bond length between single and double bonds)
 - $(4 \times 1.33 + 2 \times 1.54) \text{ \AA}$
- The correct order of increasing $\text{C} - \text{O}$ bond length of CO , CO_3^{2-} , CO_2 is
 - $\text{CO}_3^{2-} < \text{CO}_2 < \text{CO}$
 - $\text{CO}_2 < \text{CO}_3^{2-} < \text{CO}$
 - $\text{CO} < \text{CO}_3^{2-} < \text{CO}_2$
 - $\text{CO} < \text{CO}_2 < \text{CO}_3^{2-}$
- Which of the following is incorrect?
 - $\text{HI} < \text{HBr} < \text{HCl} < \text{HF}$ (ionic character)
 - $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$ (reducing strength)
 - $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$ (bond length)
 - $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$ (Stability)
- Mark out the incorrect comparison of bond angle
 - $\text{H} - \text{O} - \text{H} > \text{H} - \text{S} - \text{H}$
 - $\text{H} - \text{O} - \text{H} > \text{F} - \text{O} - \text{F}$
 - $$\begin{array}{c} \text{H} - \text{N} - \text{H} > \text{H} - \text{B} - \text{H} \\ | \qquad \qquad | \\ \text{H} \qquad \qquad \text{H} \end{array}$$
 - $$\text{O} - \text{N} - \text{O} < \text{O} = \text{N} - \text{O}$$

$$\qquad \parallel$$

$$\qquad \text{O}$$
- Consider the following hydrides and arrange them in increasing order of bond angles:

I) NH_3	II) PH_3	III) AsH_3	IV) SbH_3
a) $\text{I} < \text{II} < \text{III} < \text{IV}$	b) $\text{IV} < \text{III} < \text{II} < \text{I}$	c) $\text{I} < \text{III} < \text{II} < \text{IV}$	d) $\text{IV} < \text{II} < \text{III} < \text{I}$
- Bond energy of CH_4 is 396 k cal and that of C_2H_6 is 680 k cal . The $\text{C} - \text{C}$ bond energy in C_2H_6 in K cal is
 - 99
 - 86
 - 68
 - 120
- Which of the following boils at higher temperature.
 - CCl_4
 - CO_2
 - $\text{C}_6\text{H}_{12}\text{O}_6$
 - KCl
- The bonds present in N_2O_5 are
 - Ionic
 - Covalent
 - Ionic and covalent
 - Covalent and dative
- The bonds present in $[\text{Cu}(\text{NH}_3)_4] \text{SO}_4$ between copper and ammonia are
 - ionic
 - covalent
 - co-ordinate
 - hydrogen

11. μ of the AX_4 type of molecule is zero. The geometry of it can be:
 a) tetrahedral b) square planar c) octahedral d) any of these
12. The correct order of dipole moment is
 a) $CH_4 < NF_3 < NH_3 < H_2O$ b) $NF_3 < CH_4 < NH_3 < H_2O$
 c) $NH_3 < NF_3 < CH_4 < H_2O$ d) $H_2O < NH_3 < NF_3 < CH_4$
13. Among the following which is polar
 a) CO_2 b) SO_2 c) $BeCl_2$ d) 
14. The dipole moment of KCl is 3.336×10^{-29} Cm. The interatomic distance K^+ and Cl^- ion in KCl is 260 pm. Calculate the dipole moments of KCl, if there were opposite charges of the fundamental unit located at each nucleus. Calculate the percentage ionic character of KCl.
 a) 20 b) 40 c) 60 d) 80

More than One correct answer Type Questions

15. Which are true in terms of bond energy ?
 a) $CH_3 - F > CH_3 - Cl > CH_3 - Br > CH_3 - I$ b) $H_3C - O < H_3C - N < H_3C - C$
 c) $CH_3 - CH_2OCH_3 < CH_2 = CHOCH_3$ d) $C - N > C = N > C \equiv N$
16. Select the correct statements about the following : NO $[BF_4]$
 a) It has 5σ and 2π bonds b) It has diamagnetic character
 c) B - F bond length in this compound is shorter than that in BF_3
 d) N - O bond length in this compound is greater than that of NO.
17. What is/are true about resonance ?
 a) The resonance structures are hypothetical
 b) The unpaired electrons in various resonating structures are same
 c) Hybrid structure is most energetic
 d) Hybrid structure is least energetic
18. Which of the following statements regarding bond angles is/are true.
 a) among OSF_2 , $OSCl_2$, $OSBr_2$, halogen angle is least in OSF_2
 b) among $SbCl_3$, $SbBr_3$, SbI_3 , bond angle is least in $SbCl_3$
 c) among C_2H_2 , C_2H_4 , C_2H_6 highest bond angle is in C_2H_2
 d) the H-P-H bond angle in PH_3 is less than that in PH_4^+
19. Which of the following relation is/ are correct ?
 a) Covalent character $\propto \frac{1}{\text{Dipole moment}}$ b) Covalent $\propto$ Pseudo inert configuration
 c) Ionic character $\propto$ Inert configuration d) Ionic character $\propto \frac{1}{\text{Dipole moment}}$
20. Which of the following species will have the lone pair effect not cancelled ?
 a) ICl_2^- b) ClF_3 c) PCl_3 d) BrF_5
21. Which of the following contain co-ordinate bond?
 a) H_3O^+ b) BF_3 c) H_2O d) NH_4^+

Linked Comprehension Type Questions**Passage-I :**

In AB_n type molecules, as the electronegativity of A increases, the bond pairs come closer and the repulsion between them increases. On the other hand, as electronegativity of B increases, the bond pairs get farther and repulsion decreases.

22. Which of the following statements is true ?

- a) F-N-F angle in NF_3 is greater than H-N-H angle in NH_3
- b) F-N-F angle in NF_3 is smaller than H-N-H angle in NH_3
- c) H-O-H angle in H_2O is greater than H-N-H angle in NH_3
- d) F-O-F angle in F_2O is greater than H-O-H angle in H_2O

23. The polar as well as planar compound is

- a) H_2O_2
- b) BF_3
- c) SF_4
- d) ClF_3

24. The shape of which of the following molecules will not be distorted?

- a) BrF_3
- b) ClF_3
- c) XeF_4
- d) XeF_6

Passage-II :

Bond energy : Energy required to break a bond (equal to the energy released when a bond is formed) is named bond energy. Its unit is Kcal/mole. Stability of bond $\propto$ bond energy $\propto$ % s character

$$\alpha \frac{1}{\% \text{ p character}} \propto \text{bond order} \propto \frac{1}{\text{bond length}}$$

Higher is the number of bonds at a place, higher is the bond energy.

25. Write down bond energy order of C - H bond in (A) $-C-H$, (B) $=C-H$, (C) $\equiv C-H$

- a) $A > B > C$
- b) $A < B < C$
- c) $B > A > C$
- d) $A > C > B$

26. Write down the bond energy order of $C_2 - C_3$ in

- A) $CH_3-CH_2-CH_2-CH_3$
- B) $CH_3-CH_2-CH=CH_2$
- C) $CH_2=CH-CH=CH_2$
- D) $CH_2=CH-C\equiv CH$
- E) $CH_3-CH_2-C\equiv CH$
- F) $HC\equiv C-C\equiv CH$
- a) $E > D > C > F > B > A$
- b) $A > C > D > B > E > F$
- c) $A > C > D > E > F > B$
- d) $F > D > E > C > B > A$

Matrix Matching Type Questions

27. Column-I (Decreasing order)

- A) NH_3, SbH_3, AsH_3, PH_3
- B) HI, HBr, HCl, HF
- C) $SnH_4, GeH_4, SiH_4, CH_4$
- D) H_2O, H_2Te, H_2Se, H_2S

Column-II (Physical properties)

- P) Dipole moment
- Q) Reducing property
- R) Enthalpy of fusion
- S) Boiling point

Integer Type Questions

28. Total number of resonating structures possible for SO_4^{2-} ion is
29. The bond order of $\text{Cl}-\text{O}$ bond on ClO_3^- is 'X'. Then '3X' is:
30. The resultant dipole moment of water is 1.85D ignoring the effects of lone pair. The bond moment of each OH bond is 'X' D. Then '2X' is [given that bond angle in $\text{H}_2\text{O} = 104^\circ$, $\cos 104^\circ = -0.25$]

EXERCISE-IV

(Molecular orbital theory, Hydrogen bonding, Intermolecular forces, Metallic bond)

LEVEL-I (MAIN)Straight Objective Type Questions

- How many unpaired electrons are present in N_2^+ ?
1) 1 2) 2 3) 3 4) 4
- Which of the following has fractional bond order ?
1) O_2^{2+} 2) O_2^{2-} 3) F_2^{2-} 4) H_2^-
- The species which does not show paramagnetism is
1) O_2 2) O_2^+ 3) O_2^{2-} 4) H_2^+
- The bond order in O_2^+ is the same as in
1) N_2^- 2) CN^- 3) CO 4) NO^+
- In the formation of N_2^+ , the electron is lost from
1) a σ -orbital 2) a π -orbital 3) a σ^* -orbital 4) a π^* -orbital
- The diamagnetic molecules are
1) B_2 , C_2 , N_2 2) O_2 , N_2 , F_2 3) C_2 , N_2 , F_2 4) B_2 , O_2^{2-} , N_2
- The $\text{O}-\text{O}$ bond length in O_2 , $\text{O}_2[\text{AsF}_4]$ and $\text{K}[\text{O}_2]$ is
1) $\text{O}_2[\text{AsF}_4] < \text{O}_2 < \text{K}[\text{O}_2]$ 2) $\text{O}_2[\text{AsF}_4] < \text{K}[\text{O}_2] < \text{O}_2$
3) $\text{O}_2 < \text{O}_2[\text{AsF}_4] < \text{K}[\text{O}_2]$ 4) $\text{K}[\text{O}_2] < \text{O}_2 < \text{O}_2[\text{AsF}_4]$
- The correct stability order for N_2 and its given ions is
1) $\text{N}_2 > \text{N}_2^+ > \text{N}_2^- > \text{N}_2^{2-}$ 2) $\text{N}_2^{2-} > \text{N}_2^+ > \text{N}_2 > \text{N}_2^{2-}$
3) $\text{N}_2^+ > \text{N}_2^- > \text{N}_2 > \text{N}_2^{2-}$ 4) $\text{N}_2 > \text{N}_2^+ = \text{N}_2^- > \text{N}_2^{2-}$
- The correct bond order for CO and CO^+ are respectively
1) 3, 5/2 2) 3, 2 3) 3, 7/2 4) 5/2, 3
- During the formation of a molecular orbital from atomic orbital, the electron density is
1) minimum in nodal plane 2) maximum in nodal plane
3) zero in nodal plane 4) zero on the surface of lobe
- Bond order of NO is
1) 1.0 2) 1.5 3) 2.0 4) 2.5
- Among liq. $(\text{HF})_3$, liq NH_3 , CH_4 , CH_3OH and N_2O_4 , intermolecular hydrogen bond is expected in
1) All the molecules 2) All except one
3) Three molecules 4) Two molecules only

13. Two ice cubes are pressed over each other and unite to form one cube. Which force is responsible for holding them together ?
- Vander waal's forces
 - Covalent attraction
 - Hydrogen bond formation
 - Dipole-dipole attraction
14. The higher values of specific heat of water than other liquids has been accounted in terms of
- high dielectric constant
 - polarity
 - H-bonding
 - boiling point
15. The most suitable method of separation of a mixture of ortho and para-nitrophenol in the ratio 1:1 is
- Distillation
 - Crystallisation
 - Vaporisation
 - Colour spectrum
16. Maleic acid is stronger than fumaric acid because
- Fumaric acid shows intermolecular H-bonding
 - Fumaric acid shows intramolecular H-bonding
 - Maleic acid is dibasic acid
 - Maleic acid shows chelation

LEVEL-II (ADVANCED)

Straight Objective Type Questions

- The shape of molecular orbital depends upon
 - Size of the molecule
 - Size of the atoms involved
 - Shape of the atomic orbital
 - All the above
- The wavelength of the wave function of a bonding molecular orbital formed by LCAO is
 - Equal to the wave function of atomic orbital
 - Less than the wave function of atomic orbital
 - Greater than the wave function of atomic orbital
 - Double the wave function of atomic orbital
- The para magnetic nature of oxygen is best explained by
 - V.B. theory
 - Hybridisation
 - M.O. theory
 - VSEPR theory
- Bond order in He_2 species is
 - 0
 - 1
 - 2
 - 3
- In O_2 molecule the correct energy order of molecular orbital is
 - $\pi 2p_y > \pi 2p_z$
 - $\pi 2p_y^* = \pi 2p_z^*$
 - $\sigma 2s < \sigma 2s^*$
 - $\sigma 2s^* > \sigma 2p_x^*$
- Fractional bond order is in
 - O_2
 - O_2^+
 - O_2^{2-}
 - N_2
- Which of the following orders regarding the bond length is correct?
 - $\text{O}_2^- > \text{O}_2 > \text{O}_2^+$
 - $\text{O}_2^- < \text{O}_2 < \text{O}_2^+$
 - $\text{O}_2^- > \text{O}_2 < \text{O}_2^+$
 - $\text{O}_2^- < \text{O}_2 > \text{O}_2^+$
- Statement-1: The bond order of $\text{N}_2^+ = \text{N}_2^-$
 Statement-2: The number of e⁻s in anti bonding orbital is less in N_2^- and more in N_2^+
 - Statement-1 is True, Statement-2 is True; Statement-2 is a Correct explanation for Statement-1
 - Statement-1 is True, Statement-2 is True; Statement-2 is NOT a Correct explanation for Statement-1
 - Statement-1 is True, Statement-2 is False
 - Statement-1 is False, Statement-2 is True

9. The molecular electronic configuration of B_2 is
 a) $KK(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p)_x^1(\pi 2p)_y^1$ b) $KK(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p)_x^2$
 c) $KK(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p)^2$ d) $KK(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p)^1(\pi 2p)^1$
10. When N_2 goes to N_2^+ , the N-N bond distance ____ and when O_2 goes to O_2^+ , the O-O bond distance:
 a) increases, decreases b) decreases, increases
 c) increases, increases d) decreases, decreases
11. Which of the following species have the same bond order
 a) CN^- and CN^+ b) O_2^- and CN^-
 c) NO^+ and CN^- d) CN^- and NO^-
12. In which pair, the stronger bond is found in the first species
 A) O_2^- , O_2 B) N_2 , N_2^+ C) NO^+ , NO^-
 a) A only b) B only
 c) A and C only d) B and C only
13. The types of bonds present in $CuSO_4 \cdot 5H_2O$ (blue vitriol) are
 a) electrovalent and covalent
 b) electrovalent, covalent, co-ordinate and hydrogen bond
 c) covalent and co-ordinate covalent
 d) electrovalent
14. Inter molecular hydrogen bonding is absent in
 a) H_2O b) NH_3 c) C_2H_5OH d) CH_4
15. Which among the following has the highest volatility
 a) H_2O b) H_2S c) H_2Se d) H_2Te
16. Strongest hydrogen bonds can be formed by
 a) HF b) H_2O c) NH_3 d) HCl
17. **LIST-1** **LIST-2**
 A) NaCl 1) Covalent bond
 B) CH_4 2) Ionic bond
 C) NH_4^+ 3) Metallic bond
 D) Cu metal 4) Covalent and dative bond
- The correct match is
- | | A | B | C | D |
|----|---|---|---|---|
| a) | 2 | 1 | 4 | 3 |
| c) | 2 | 3 | 1 | 4 |
- | | A | B | C | D |
|----|---|---|---|---|
| b) | 2 | 4 | 1 | 3 |
| d) | 1 | 2 | 3 | 4 |
18. (A) : Water is a liquid whereas sulphurdioxide is a gas at room temp
 (R) : Molecular mass of SO_2 is more than that of H_2O
 a) Both (A) and (R) are true and (R) is the correct explanation of (A)
 b) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 c) (A) is true but (R) is false
 d) (A) is false but (R) is true

19. (A) : O-Hydroxy benzaldehyde is steam volatile but not P-hydroxy benzaldehyde
 (R) : Intramolecular hydrogen bond is present in orthohydroxy benzaldehyde but intermolecular hydrogen bond in parahydroxy benzaldehyde
- a) Both (A) and (R) are true and (R) is the correct explanation of (A)
 b) Both (A) and (R) are true and (R) is not the correct explanation of (A)
 c) (A) is true but (R) is false
 d) (A) is false but (R) is true
20. In solid argon, the atoms are held together by
- a) Ionic bonds
 b) Hydrogen bonds
 c) Vander Waal forces
 d) Hydrophobic forces

More than One correct answer Type Questions

21. Which of the following elucidate molecular orbital theory rightly ?
- a) Number of MO's are equal to the number of AO's from which they are formed
 b) BOM's increase electron density between the nuclei
 c) ABMO's have a region of zero electron density
 d) Number of electrons in BMO's are more than in ABMO's
22. Which of the following statements is/are true?
- a) O_2 is paramagnetic and O_3 is diamagnetic
 b) O_2 and B_2 both are paramagnetic
 c) B_2 and C_2 both are paramagnetic
 d) O_2 is paramagnetic but B_2 is diamagnetic
23. Which of the following species have bond order equal to three?
- a) CN^-
 b) CO
 c) NO^+
 d) O_2^+
24. In which of the following sets all the species are paramagnetic in nature ?
- a) O_2, O_2^{2+}, N_2^{2-}
 b) B_2, C_2, H_2
 c) O_2^-, O_2^+, O_2
 d) N_2^+, O_2^+, F_2^+
25. Which of the following statement(s) is/are incorrect ?
- a) O_2 is paramagnetic, O_3 is also paramagnetic
 b) O_2 is paramagnetic, O_3 is diamagnetic
 c) B_2 is paramagnetic C_2 is also paramagnetic
 d) Different observation is found in their bond length when $NO \rightarrow NO^+$ and $CO \rightarrow CO^+$
26. Incorrect option (s) is (are)
- a) $NaCl$ is soluble in CCl_4
 b) KHF_2 has ionic bond
 c) The shape of ClF_3 is T-shape
 d) $LiCl$ is completely ionic compound
27. Which of the following molecules involve intramolecular H-bond?
- a) Ortho-nitrophenol
 b) Chloral hydrate
 c) Para-nitrophenol
 d) Benzoic acid

Linked Comprehension Type Questions

Passage-I :

In poly atomic molecules the bond energy of similar bond may differ due to the difference in the chemical environmental environment around central atom. Bond order is given by the number of bonds between the two atoms in a molecule. Iso electronic molecules and ions will have identical bond orders. In general as bond order increases the bond energy increases and bond length decreases.

28. Among the following degenerate orbitals are
 a) $\sigma 1s, \sigma 1s^*$ b) $\sigma 2px, \sigma 2px^*$ c) $\pi 2p_x, \pi 2p_y$ d) $\pi 2p_y, \pi 2p_z^*$
29. Which of the following shows maximum bond order
 a) H_2 b) N_2 c) O_2 d) F_2
30. Identify the pair which shows similar bond order ?
 a) O_2^-, O_2^{2-} b) O_2^-, N_2 c) CO, NO^+ d) F_2, O_2

Passage-II :

The electrons in atoms occupy atomic orbitals (AO_s) that are represented as the regions around the nuclei where there is a high probability of finding the electrons. In the so-called LCAO (linear combination) approach, as pioneered by Hund and Mulliken, when AOs come close together, they overlap forming MOs (molecular orbitals). Two AOs can overlap to form two MOs, one of which lies at a lower energy level (BMO) than the other at a higher energy level and is called an antibonding molecular orbital (ABMO). Each MO can hold one or two electrons in accordance with Pauli's exclusion principle. MOT can explain the paramagnetism of molecules such as O_2 and NO and other spectral features.

31. In a molecule number of electrons in bonding MO is more as compared to antibonding MO, hence
 a) a bond will be formed b) no bond will be formed
 c) information is not sufficient d) none of the above
32. Bond strength increases with
 a) bond length increasing b) antibonding electrons being higher in number
 c) bond order increasing d) bond angle increasing
33. Molecular orbital theory is preferred over valence bond theory as it can best explain
 a) bond strength b) geometry of molecule
 c) diamagnetism in molecules d) paramagnetism in molecules

Matrix Matching Type Questions

- | 34. Column-I | Column-II |
|--------------------------|------------|
| A) Fractional bond order | P) O_2^- |
| B) Paramagnetic nature | Q) O_2^+ |
| C) Bond order value 3 | R) NO^+ |
| D) Bond order value 2.5 | S) CN |

Integer Type Questions

35. The number of non-bonding electrons of N_2 is
36. The number of π bonds present in C_2 molecule is
37. The π^* molecular orbital has _____ nodal planes.
38. The maximum number of hydrogen bonds in which a molecule can participate in water is
39. The maximum possible number of hydrogen bonds in which a water molecule can participate in ice is
40. The number of completely filled π molecular orbitals in ground state C_6H_6 is

KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL-I

1) 4 2) 1 3) 3 4) 2 5) 3 6) 2 7) 1 8) 1

9) 4 10) 2 11) 1 12) 4 13) 2 14) 3 15) 2

LEVEL-II

1) a 2) c 3) b 4) c 5) b 6) d 7) a 8) d

9) c 10) c 11) b 12) d 13) d 14) b 15) d 16) a

17) c 18) a 19) c 20) c 21) ab 22) ad 23) a 24) ad

25) a 26) c 27) b 28) d 29) b 30) d 31) a 32) d

33) A-Q; B-S; C-R; D-P 34) 3 35) 7 36) 0 37) 8 38) 4

EXERCISE-II

LEVEL-I

1) 2 2) 1 3) 3 4) 4 5) 3 6) 4 7) 1 8) 2

9) 2 10) 2 11) 4 12) 1 13) 3 14) 3 15) 3

LEVEL-II

1) a 2) b 3) a 4) c 5) a 6) b 7) a 8) b

9) c 10) c 11) a 12) a 13) c 14) a 15) a 16) b

17) a 18) ac 19) abcd 20) abc 21) bcd 22) ab 23) bcd 24) acd

25) cd 26) ab 27) ac 28) c 29) a 30) c 31) b 32) c

33) c 34) A-QR; B-S; C-P; D-P 35) 0 36) 7 37) 4 38) 8

39) 3 40) 2 41) 3 42) 3 43) 4 44) 2

EXERCISE-III

LEVEL-I

1) 1 2) 4 3) 2 4) 1 5) 1 6) 1 7) 2 8) 1

9) 2 10) 3 11) 2 12) 1 13) 2 14) 3 15) 2 16) 4

17) 1 18) 1 19) 2 20) 1 21) 3

LEVEL-II

1) c 2) c 3) d 4) d 5) c 6) b 7) b 8) d

9) d 10) c 11) d 12) a 13) b 14) d 15) abc 16) ab

17) abd 18) abcd 19) abc 20) bcd 21) ad 22) b 23) d 24) c

25) b 26) d 27) A-PR; B-Q; C-QRS; D-SR 28) 6 29) 5

30) 3

EXERCISE-IV

LEVEL-I

1) 1 2) 4 3) 3 4) 1 5) 1 6) 3 7) 1 8) 1

9) 3 10) 3 11) 4 12) 3 13) 3 14) 3 15) 1 16) 1

LEVEL-II

1) c 2) a 3) c 4) a 5) c 6) b 7) a 8) c

9) a 10) a 11) c 12) d 13) b 14) d 15) b 16) a

17) a 18) b 19) a 20) c 21) abc 22) ab 23) abc 24) cd

25) acd 26) ad 27) ab 28) c 29) b 30) c 31) a 32) c

33) d 34) A-PQS; B-PQS; C-R; D-S 35) 4 36) 2 37) 2

38) 4 39) 3 40) 3